Indian Institute of Technology Guwahati

Department of Physics

PH102

Tutorial-3

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1. Convert points $P(1,3,5)$, $T(0,-4,3)$ and $S(-3,-4,-10)$ from Cartesian to cylindrical and spherical coordinates.

2. A vector field is given by

$$\vec{Q} = \frac{\sqrt{x^2+y^2+z^2}}{\sqrt{x^2+y^2}}[\hat{x}(x-y)+\hat{y}(x+y)]$$

   Evaluate $\vec{\nabla}\times\vec{Q}$ and $\vec{\nabla}\cdot\vec{Q}$ in

   (a) cylindrical polar coordinate system,

   (b) spherical polar coordinate system.

3. (a)Find the Laplacian of the scalar field $U = x^2y + xyz$

   (b)Find the Laplacian of the vector field $\vec{A} = 2s\cos\phi\hat{s} - 4s\sin\phi\hat{\phi} + 3\hat{z}$

4. Compute the divergence of the function $\vec{V} = (r\cos\theta)\hat{r} + (r\sin\theta)\hat{\theta} + (r\sin\theta\cos\phi)\hat{\phi}$. Check the divergence theorem for this function, using as your volume the inverted hemispherical bowl of radius R, resting on the xy plane and centered at the origin (as shown in figure 1).

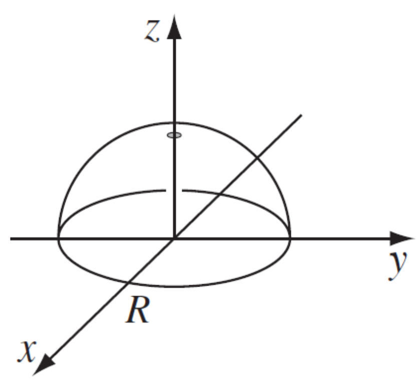

Figure 1

5. Find the divergence of the function

   (a) $\vec{V} = s(2 + \sin^2\phi)\hat{s} + s\sin\phi\cos\phi\hat{\phi} + 3z\hat{z}$

   (b) Find the curl of $\vec{V}$

6. Evaluate the following integrals

   (a) $\int_0^5 \cos x\,\delta(x-\pi)dx$

   (b) $\int_{-\infty}^{\infty} \ln(x+3)\,\delta(x+2)dx$

   (c) $\int_{-1}^{1} 9x^2\,\delta(3x+1)dx$

   (d) $\int_{-\infty}^{a} \delta(x-b)dx$

   (e) $J = \int_v e^{-r}(\vec{\nabla}\cdot\frac{\hat{r}}{r^2})d\tau$